SourceTrail Architecture Overview

SourceTrail is a multi-source retrieval-augmented generation (RAG) knowledge base. Visitors upload PDF, DOCX, Markdown, TXT and CSV files or add web pages, then ask questions and receive grounded answers with inline citations such as [1] and [2].

Every citation can be inspected: clicking it shows the document, page or section, and the exact chunk of text the answer relied on. Each chunk has a stable identifier built from the document slug, the page number and the chunk position, for example annual-report-2024:p47:c003.

The system runs entirely on free infrastructure: a FastAPI service, PostgreSQL with the pgvector extension, CPU-only ONNX models from FastEmbed, and a hosted open-weight LLM.

Ingestion pipeline

Each file type has its own extractor. PDFs are split into one section per page, Markdown and Word documents into one section per heading, and CSV files into one section per row. Web pages are fetched with strict safety checks and their main article text is extracted with trafilatura, which removes menus and footers.

Sections are cut into chunks of about 1,000 characters with an overlap of 150 characters, preferring paragraph, then line, then sentence boundaries. CSV rows are grouped into chunks but a single row is never split.

Chunks are embedded with the bge-small-en-v1.5 model, which produces 384-dimensional vectors. The document title and section heading are prepended before embedding so that each vector carries its context. Storage happens in a single transaction: a document is saved with all of its chunks or not at all.

Retrieval and answering

Every question runs two searches. Vector search finds the 20 chunks closest in meaning using cosine distance on an HNSW index. Keyword search finds the 20 best word matches using PostgreSQL full-text search ranked with ts_rank_cd.

The two ranked lists are merged with Reciprocal Rank Fusion. Each chunk receives 1 / (60 + rank) points from every list it appears in, so chunks that both methods agree on rise to the top, and the different score scales of the two methods do not matter.

The 30 best fused candidates are re-scored by a cross-encoder, ms-marco-MiniLM-L-6-v2, which reads the question and each chunk together. The top 6 chunks are sent to the LLM as numbered sources. Citation markers that point to sources that were not provided are removed before the answer is shown.

Safety on a public site

SourceTrail has no logins. Each visitor receives an anonymous workspace identified by a random ID stored in an HTTP-only cookie. Searches cover the visitor's own workspace plus the shared demo collection, and no visitor can see another visitor's documents.

Rate limits are keyed by a hash of the client IP address: 10 questions per minute and 20 uploads per hour. Each workspace may hold 10 documents and 2,000 chunks. A site-wide cap of 800 AI answers per day keeps usage within the free LLM quota; when it is reached, visitors still see the most relevant passages.

Visitor workspaces are deleted automatically 24 hours after their last activity. The system prompt instructs the model to treat uploaded documents as information and never as instructions, which defends against prompt injection.